

653 notes

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1 $\pi - \lambda$ Theorem

Claim: Let $\mathcal{C}$ be a π -system. Then $\lambda(\mathcal{C}) = \sigma(\mathcal{C})$.

Proof: Let $\{\Lambda_\alpha\}$ be a collection of λ -systems. I claim that $\Lambda = \bigcap_\alpha \Lambda_\alpha$ is a λ -system. Indeed, $\emptyset \in \Lambda_\alpha \rightarrow \emptyset \in \Lambda$. If $A \in \Lambda$, then $A^c \in \Lambda_\alpha \rightarrow A^c \in \Lambda$. Λ is closed under countable disjoint union.

Every σ -algebra is a λ -system, so $\lambda(\mathcal{C}) \geq \sigma(\mathcal{C})$. Define $\lambda_1 = \{A \leq X : A \cap C \in \lambda(\mathcal{C}) \text{ for all } C \in \mathcal{C}\}$. We have $\emptyset \in \lambda_1$. λ_1 is closed under complements since $A^c \cap C = C - A \cap C$. Let $\{A_n\} \leq \lambda_1$ be disjoint. Then

$$C \cap \bigcup_{n \in \mathbb{Z}^+} A_n = \bigcup_{n \in \mathbb{Z}^+} C \cap A_n \in \lambda(\mathcal{C})$$

as a countable disjoint union of elements in $\lambda(\mathcal{C})$.